



ASSIGNMENT 3

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Course : GAME PROGRAMMING

SECTION : BS SE 5 A

Date : 6 June , 2026

Fall Guys — Code Architecture Design Document

1. Game Overview

Fall Guys is a massively multiplayer party game developed by Mediatonic. Up to **60 players** compete across chaotic obstacle course rounds. Each round eliminates a portion of players until a single winner remains in the final round. The game uses physics-based movement, real-time collision, and round-based elimination — making its architecture significantly more complex than Among Us.

Key differences from Among Us:

- 60 players simultaneously (vs 15)
- Physics engine heavily involved (ragdoll, collision)
- No social deduction — pure real-time competition
- Multiple rounds with different game modes per match
- Dedicated servers (not host-client)

2. High-Level Architecture

Fall Guys uses a **Dedicated Server Architecture** — no player is the host. A central game server runs the authoritative physics and game logic.

Major Systems:

System	Role
Player System	Movement, physics, costume, elimination state
Matchmaking System	60-player lobby, region, skill-based grouping
Round Manager	Round selection, loading, transition between rounds
Physics System	Server-side collision, ragdoll, obstacle movement
Elimination System	Tracks who qualifies/gets eliminated each round
Game Mode System	Race, Survival, Team, Final — different rules per round
Network Sync System	Position sync for 60 players simultaneously

System	Role
UI System	HUD, elimination feed, countdown, crown display
Cosmetics System	Costume loading, emotes, nameplates

3. Code Structure

3.1 Folder Structure

Assets/Scripts/

Core/

GameManager.cs

RoundManager.cs

GameStateMachine.cs

Player/

PlayerController.cs

PlayerPhysics.cs

PlayerAnimator.cs

CostumeManager.cs

Systems/

Matchmaking/

LobbyManager.cs

QueueManager.cs

Rounds/

BaseRound.cs

RaceRound.cs

SurvivalRound.cs

TeamRound.cs

FinalRound.cs

Elimination/

EliminationManager.cs

QualifyZone.cs

FallZone.cs

Physics/

ObstacleController.cs

MovingPlatform.cs

RagdollHandler.cs

Network/

ServerNetworkManager.cs

PlayerStateSyncer.cs

RPCHandler.cs

UI/

HUDManager.cs

EliminationFeedUI.cs

RoundTransitionUI.cs

LeaderboardUI.cs

3.2 Core Classes

GameManager — Singleton

GamePhase currentPhase

int totalPlayersAlive

List<PlayerController> allPlayers

void ChangePhase(GamePhase phase)

void CheckRoundEnd()

PlayerController — MonoBehaviour

string playerName

bool isEliminated

bool hasQualified

Vector3 position, velocity

void Move(Vector2 input)

void OnFallOutOfMap()

void OnReachQualifyZone()

RoundManager — Singleton

List<BaseRound> roundQueue

BaseRound currentRound

int currentRoundNumber

void LoadNextRound()

void OnRoundComplete()

void SelectRoundType(int playerCount)

BaseRound — Abstract Class

int qualifyCount

float roundTimeLimit

abstract void OnRoundStart()

abstract void OnRoundEnd()

abstract void CheckWinCondition()

(RaceRound, SurvivalRound, TeamRound, FinalRound extend this)

EliminationManager — Singleton

List<PlayerController> eliminatedPlayers

List<PlayerController> qualifiedPlayers

void EliminatePlayer(PlayerController p)

void QualifyPlayer(PlayerController p)

bool IsRoundOver()

3.3 Game State Machine

MATCHMAKING → LOBBY → ROUND_LOADING → IN_ROUND →

ROUND_RESULT → ELIMINATION_SCREEN → ROUND_LOADING (next round)

→ FINAL_WINNER (if last round)

3.4 Game Mode Types

Mode	Example Rounds	Win Condition
Race	Dizzy Heights, Slime Climb	First N players to finish
Survival	Jump Club, Roll Out	Last N players standing
Team	Egg Scramble, Rock N Roll	Winning team qualifies
Final	Crown Climb, Fall Mountain	Single winner takes crown

4. Design Decisions

1. Dedicated Server (not Host-Client) 60 players ka physics ek player ki machine pe run karna impossible hai. Dedicated server authoritative physics calculate karta hai — sab clients sirf input bhejte hain aur server se positions receive karte hain.

2. Abstract BaseRound Class (Strategy Pattern) Har round ka alag logic hai — Race alag, Survival alag, Final alag. BaseRound abstract class banane se naya round add karna sirf ek naya class extend karna hai. GameManager ko koi change nahi chahiye.

3. Server-Side Physics Cheating rokne ke liye physics server pe run hoti hai. Client sirf W/A/S/D input bhejta hai, server calculate karta hai actual position. Laggy but fair for all 60 players.

4. Component-Based Player PlayerController, PlayerPhysics, PlayerAnimator alag components hain taake Ghost/Eliminated players mein sirf Physics disable ho — baki sab reuse ho.

5. Event Bus for Elimination EliminationManager OnPlayerEliminated event broadcast karta hai — UI, Sound, Network sab subscribe karte hain. Tight coupling nahi.